Prodigy Anesthesia Quick Review

Ultrasound-Guided Nerve Blocks

What are the frequencies in which a medical ultrasound machine functions?	2-13 MHz *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
Are there size limits to structures identified by an ultrasound?	Yes, an ultrasound beam has as an average wavelength of less than 1mm, thus limiting ultrasound usage to identifying anatomical structures 1 mm in diameter or greater. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
What type of ultrasound probes (higher-frequency or lower-frequency) would be best to visualize superfical structures that are less than 4cm beneath the skin?	High-frequency ultrasound probes *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
What determines the amount of ultrasound waves reflected by a tissue back to the probe?	It is determined the acoustic impedance of the tissue, which is the degree of resistance the tissue has to the passage of acoustic waves. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
How does tissue that is a poor reflector of ultrasound waves appear on an ultrasound display monitor?	It appears as shades of grey (hypoechoic). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
How does tissue that reflects large amounts of ultrasound waves appear on an ultrasound display monitor?	It appears white (hyperechoic). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
1. If an area on an ultrasound monitor is referred to as anechoic, what does that mean? 2. What color will it appear on the monitor?	Areas referred to as anechoic, lack the ability to relect ultrasound waves. 2. Black *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
1. What does the vertical axis of an ultrasound monitor represent? 2. What does the horizontal axis display?	1. The distance from the probe to the block needle or to anatomical structures 2. The length to the left or right of the probe's center *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*

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Why would an in-plane (axial/longitudinal) approach to an ultrasound-guided regional anesthetic be advantageous over an out-of-plane (tangential/short axis) approach?	Using an in-plane approach, the entire length of the needle is visible on the ultrasound screen. Using an out-of-plane approach, only a cross section of the needle is displayed, making interpretation of its position more difficult. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1107.*
List advantages of ultrasound-guided regional anesthetics?	Direct visualization of the spread of local anesthetic, direct visualization of anatomy, better performance, faster onset times, allows for more accurate placement and decreased amounts of local anesthetics, increased patient satisfaction, improved quality of block, administration that is less painful *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1106.*
How are patients positioned for an ultrasound-guided interscalene block?	Patients are positioned supine with the head turned away from the operative shoulder. The ultrasound probe is then directed toward the level of the first rib. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
Is a high-frequency or low-frequency ultrasound probe used for an interscalene block?	High-frequency *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1109.*
When performing an ultrasound-guided interscalene block, what two anatomical structures should be visualized first?	The first rib and the subclavian artery are first identified. The probe is then moved superiorly from the first rib to the level (or slightly below) of the cricoid cartilage. Here, the brachial plexus appears as a collection of dark circular structures that are located between the anterior and middle scalene muscles. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1109.*
Where are the 3 roots of the brachial plexus located in relation to the subclavian artery?	They will be found lateral to the subclavian artery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1109.*
Once the brachial plexus has been identified, where is the needle inserted for an ultrasound-guided interscalene block?	In-plane with the ultrasound probe, a 2-inch, 22-gauge needle is placed behind the brachial plexus with the needle tip stopping very close to the roots. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1109.*
How much local anesthetic is injected for an ultrasound-guided interscalene block? How will this appear on the ultrasound monitor?	Approximately 20 mls are injected under direct visualization. On the monitor, the spread of local anesthetic will produce a hypoechoic area surrounding the plexus. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1109.*

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What two muscles form the crease in the axilla in which the ultrasound probe is placed during an ultrasound-guided axillary block?	Biceps and pectoralis major *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
What two anatomical landmarks need to be identified first in order to perform an ultrasound-guided axillary nerve block?	The brachial plexus terminal nerve branches and the axillary artery *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
Can the median nerve be visualized during an ultrasound-guided axillary nerve block?	Yes. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
1. Where is the ulnar nerve located in relation to the axillary artery? 2. Where is the radial nerve in relation to the axillary artery?	1. The ulnar nerve is medial to the axillary artery. 2. The radial nerve is inferior to the axillary artery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
Should the ultrasound beam be directed parallel or perpendicular to the axillary artery?	Perpendicular *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
Once landmarks have been directly visualized, where is the needle inserted for an axillary nerve block?	Using an in-plane technique, a 22-gauge, 2-inch needle is placed anterior to the brachial plexus. It is inserted until the tip of the needle lies very near the terminal branches. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
How much local anesthetic is used for an ultrasound-guided axillary nerve block?	approximately 20-30 ml *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
Is the musculocutaneous nerve typically blocked during an axillary nerve block?	No. In the axilla, the musculocutaneous nerve is typically separated from the ulnar, median, and radial nerves, and therefore must be blocked individually. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
The musculocutaneous nerve is situated between what two muscles?	Pectoralis major and coracobrachialis muscle *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*

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How is the patient positioned for an ultrasound-guided femoral nerve block?	The patient is supine with the inguinal area exposed. If a nerve stimulator is used in conjunction with the ultrasound, the thigh and patella should be exposed as well. *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 399.*
How much local anesthetic is injected during an ultrasound-guided musculocutaneous nerve block?	5-10 ml *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1113.*
In the groin, where is the femoral nerve in relation to the femoral artery?	Once the femoral nerve enters the groin it is positioned lateral to the femoral artery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1119-1120.*
What lumbar nerve roots give rise to the femoral nerve?	L2-L4 nerve roots *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1119.*
1. Where is the ultrasound probe placed when performing a femoral nerve block? 2. What are the 3 landmarks to visualize?	1. The probe is placed transversely in the inguinal crease, parallel to the inguinal ligament. 2. The femoral artery, femoral vein, and the femoral nerve *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1121.* *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 399.*
How will the femoral nerve appear on the ultrasound monitor?	Hyperechoic, triangular, and lateral to the femoral artery *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1120.*
Once the femoral nerve is visualized, where is the needle inserted for an ultrasound-guided femoral nerve block?	In-plane to the ultrasound beam, the 2-inch, 22-gauge stimulating needle is inserted lateral to the identified nerve, through the fascia iliaca, and advanced toward the nerve. Providers can frequently feel a "pop" sensation as the needle passes through the fascia iliaca. *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 400.* *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1120.*
How much local anesthetic is used for an ultrasound-guided femoral nerve block?	20 ml *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1120.*

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As the sciatic nerve approaches the popliteal fossa it splits into two branches. What are they?	1. common peroneal nerve 2. tibial nerve *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1122.*
Where is the ultrasound probe initially placed for a popliteal sciatic block?	1. Transversely in the popliteal crease *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1122.*
1. What anatomical structures are visualized when the ultrasound probe is placed in the popliteal crease?	The popliteal artery, popliteal vein, and typically the tibial nerve are found at the level of the popliteal crease. The tibial nerve is superficial and slightly lateral to the popliteal vessels and is flanked laterally by the biceps femoris muscle and medially by the semimembranosus and semitendinosus muscles. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1122.* *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 442.*
Once the tibial nerve has been identified, what is the next step in an ultrasound-guided popliteal sciatic block?	Once the tibial nerve has been visualized, the probe is moved proximally until the common peroneal nerve can be seen joining the tibial nerve to form the sciatic nerve. This union generally occurs approximately 5-10 cm above the popliteal crease. In plane to the ultrasound beam, the needle is inserted lateral to the sciatic nerve and advanced toward the nerve. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1122.* *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 444.*
How much local anesthetic is used in an ultrasound-guided popliteal sciatic block?	20 ml *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1122.*
What nerve segments are blocked by an ultrasound-guided transversus abdominis plane block?	T9-L1 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1125.*
Superficial to deep, list the abdominal muscles that are located between the subcostal margin and iliac crest.	External oblique, internal oblique, transversus abdominis *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1125.*

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An ultrasound-guided transversus abdominis plane block is placed in the lateral abdominal wall between what two muscle layers?	It is placed between the internal oblique and the transversus abdominis muscles. Through a single injection, nerves that lie within this plane (T9-L1), are blocked unilaterally. The block must be performed on both sides of the abdomen for bilateral anesthesia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1125.*
Where is the ultrasound transducer placed when performing a transversus abdominis plane block?	With the patient supine, the probe is placed transversely between the anterior superior iliac spine and the 12th rib. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1125.*
Once the ultrasound probe is in place, and structures have been identified, how is a transversus abdominis plane block performed?	In plane to the ultrasound beam, a 21 gauge 4 inch needle is placed through the muscle layers. The end of the needle should rest between the internal oblique and transversus abdominis muscles. 20 ml is injected per side in an adult patient. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1125.* *Hadzic A, Carrera A, Clark T, et al. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 462.*

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